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LAB07: Tiling on PULP

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Francesco Conti – f.conti@unibo.it

Objective of the Class

Intro: Tiling

Tasks:

- 2D convolution in L1
- 2D convolution in L2
- Layer Tiling

Deadline:

Dec 5th 2025

Programming Language: C

Lab duration: 3h

Assignment:

- Time for delivery: 2 weeks

<u>The class is meant to be interactive: coding together and on your own!</u>

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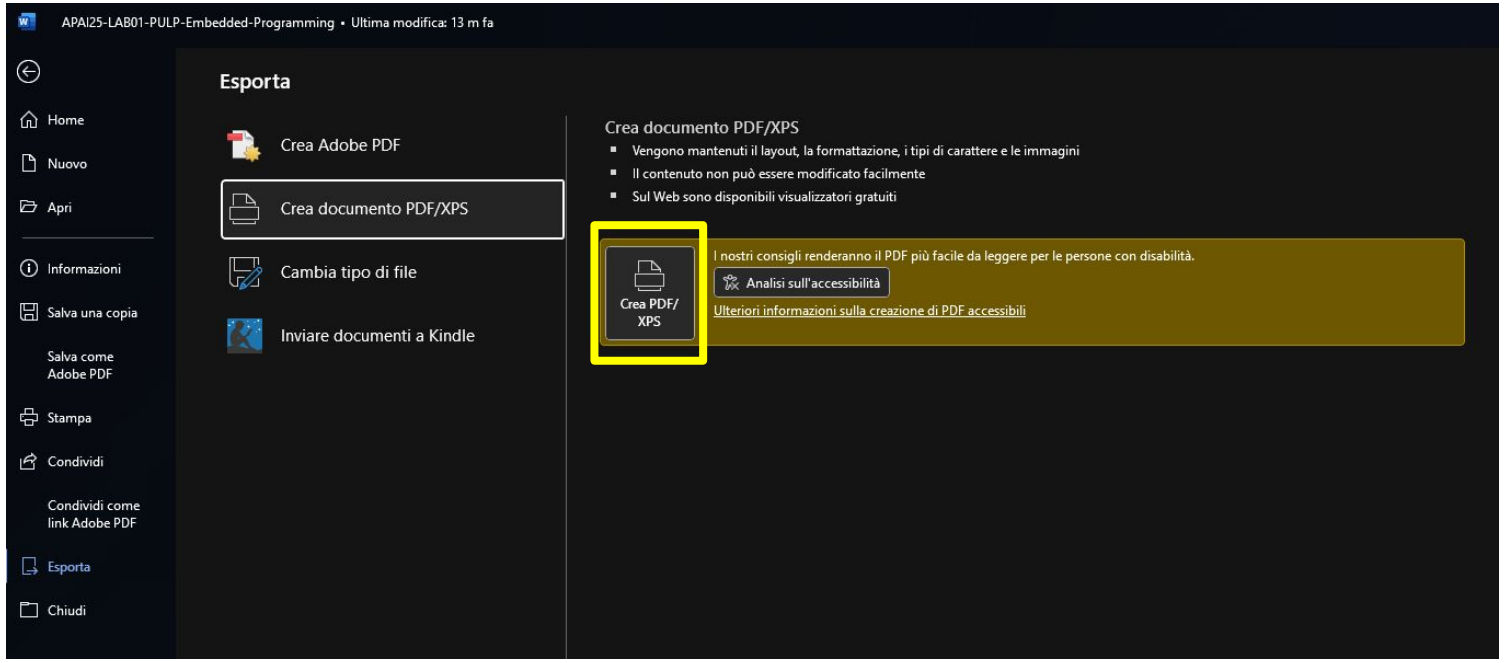
How to deliver the Assignment

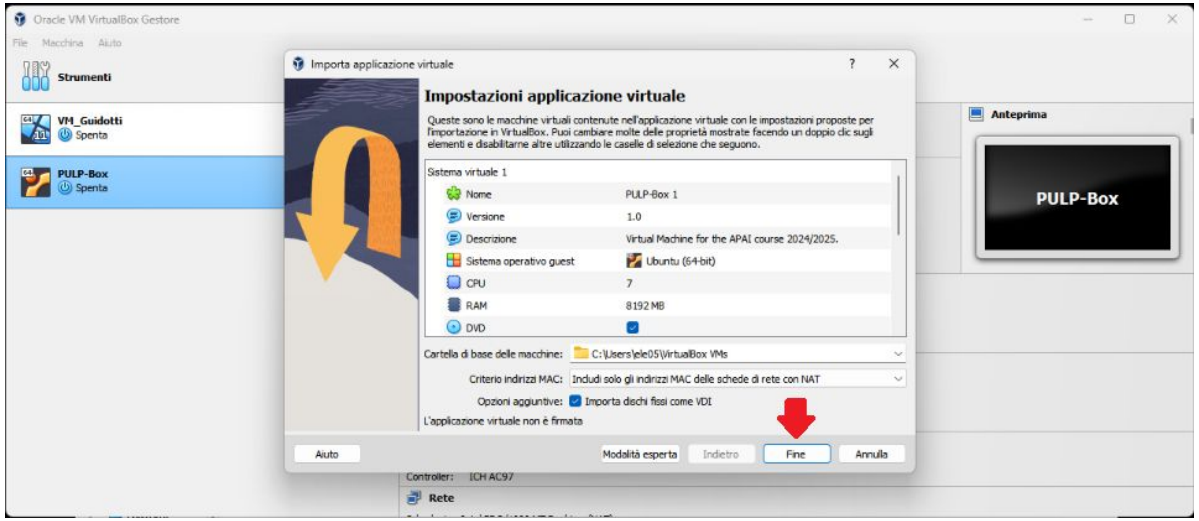
You will deliver ONLY the PDF assignment, no code

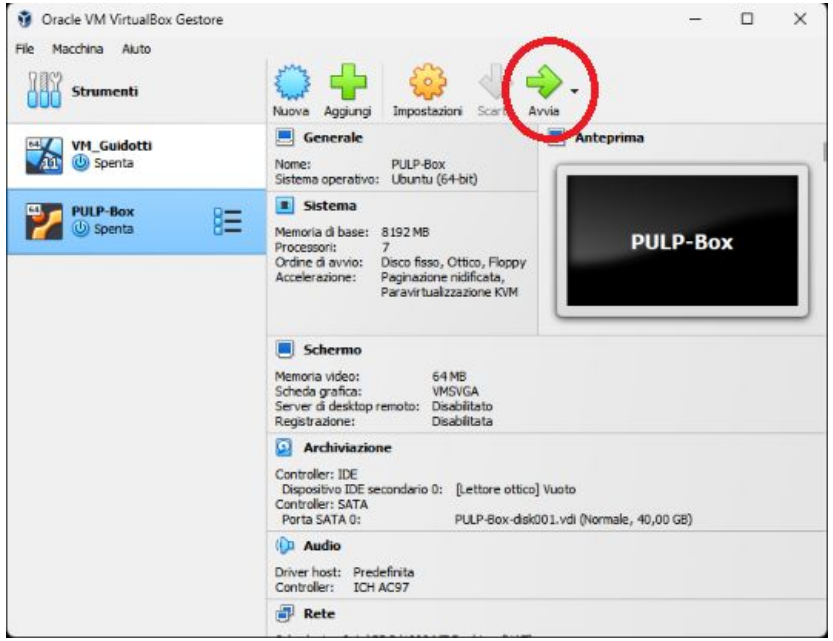
- Download the assignment file from Virtuale.
- Fill the results required by the assignment.
- Export to pdf format.
- Rename the file to: LAB<number_of_the_lesson>_APAI_<your_name>.pdf
- Use Virtuale platform to load ONLY your .pdf file



Opening the VM and VSCode

1. On the lab's PCs, open the file explorer and go to This PC, C:/VM_APAI/
2. Double click on PULP-box.ova
3. VirtualBox opens, just click on "Fine"
4. Wait for the VM to be imported
5. Open the VM with "Avvia"





Password is 'pulp'

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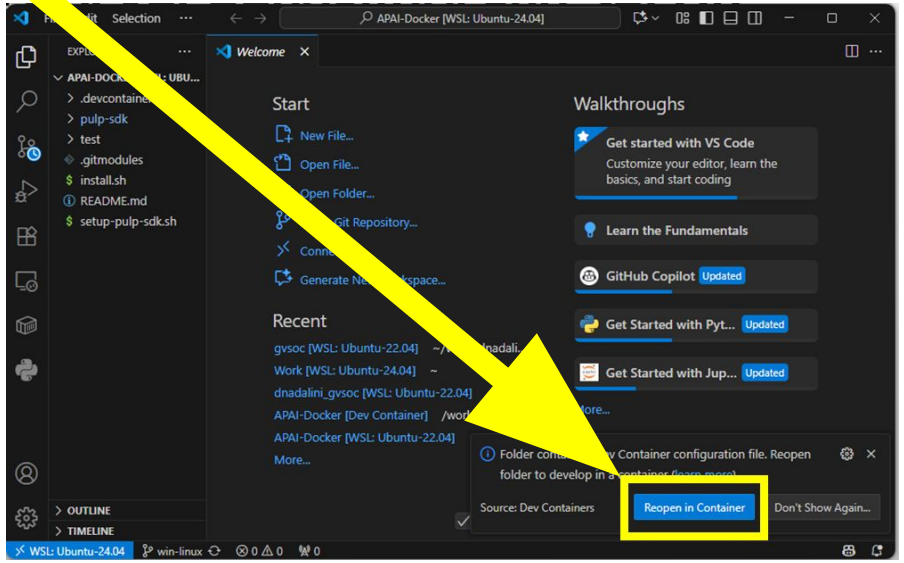
Opening the Docker with VSCode

1. Open a terminal (right click – open a new terminal)
2. From the terminal, open VSCode in the folder of the Docker

```
$ cd APAI-Docker
$ code .
```

3. Reopen the APAI-Docker folder in VSCode (click on “Reopen in container”)
4. Now you can use the integrated terminal (open with CTRL+J) to run your applications!



IMPORTANT: every time you open a new terminal to work on PULP, launch

```
$ source setup-pulp-sdk.sh
```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Getting Started:

IMPORTANT: activate the pulp-sdk module file <u>every</u> time a new shell is open.

```
$ source setup-pulp-sdk.sh
```

HOW TO RUN THE CODE:

```
$ git clone https://github.com/EEESlab/APAI25-LAB07-PULP-Tiling-Part1
$ cd APAI25-LAB07-PULP-Tiling-Part1
$ make clean all run
```

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INTRO

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TASK1: fit in L1

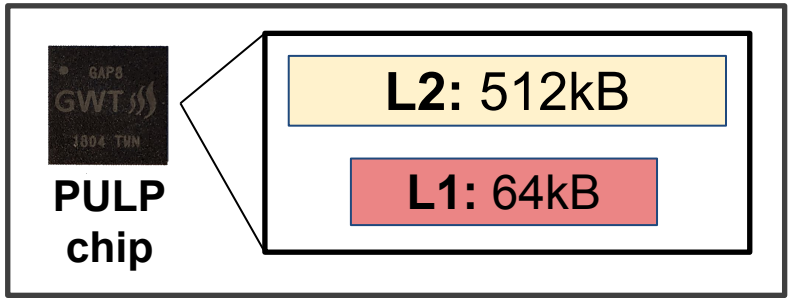
Case study: 1x1 conv2D

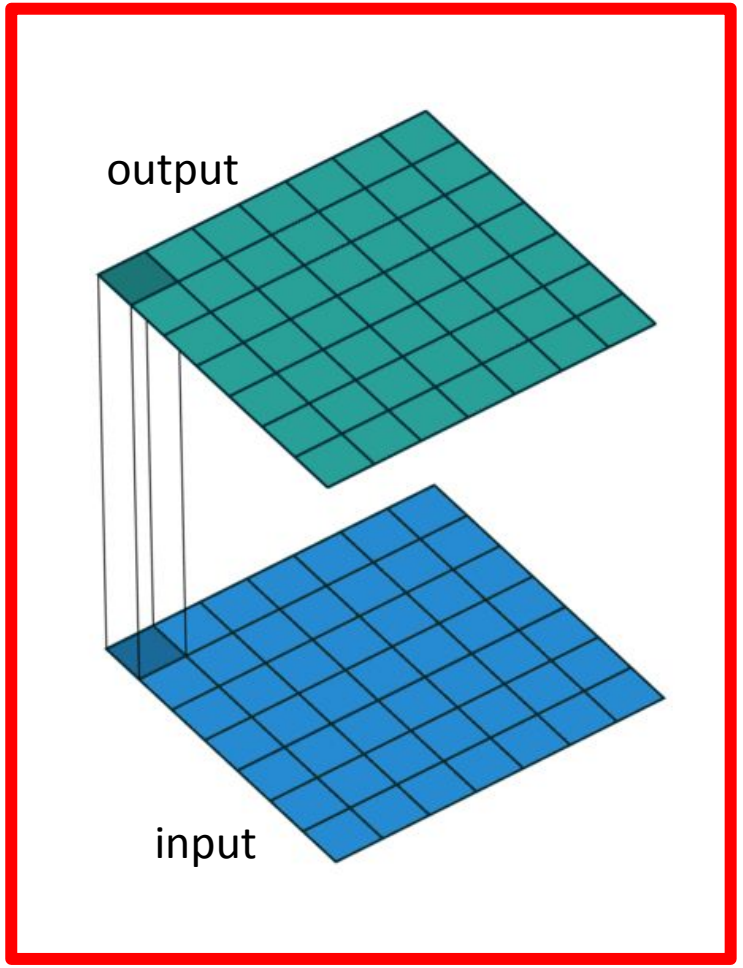
We tackle a 1x1 convolution with this sizes:

- Input = SPATIAL_DIM → defined by you
- Output = SPATIAL_DIM → defined by you
- Kernel = 1x1
- Stride = 1
- Padding = 0

NB: with conv1x1 the spatial size between input and output does not change!

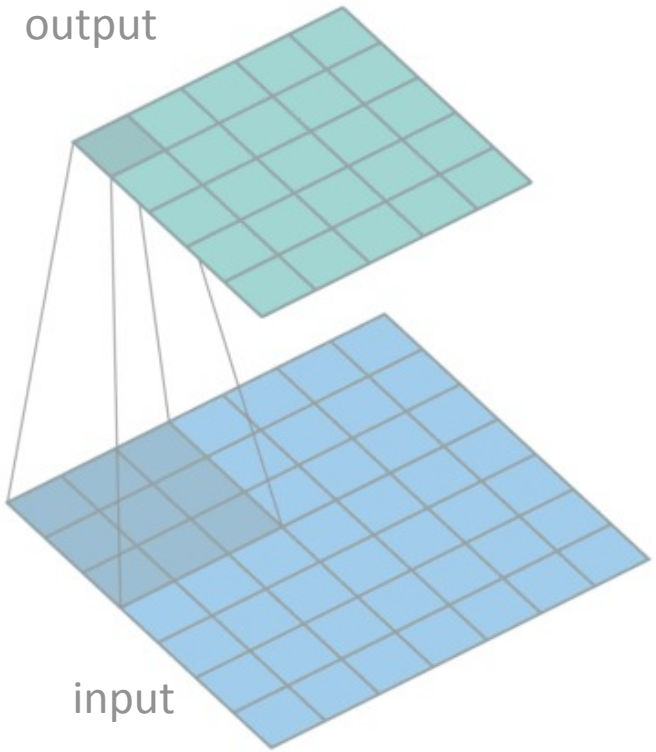
We want to fit into the L1 memory!





1x1 Convolution

Used today!



3x3 Convolution

Used in lab04!

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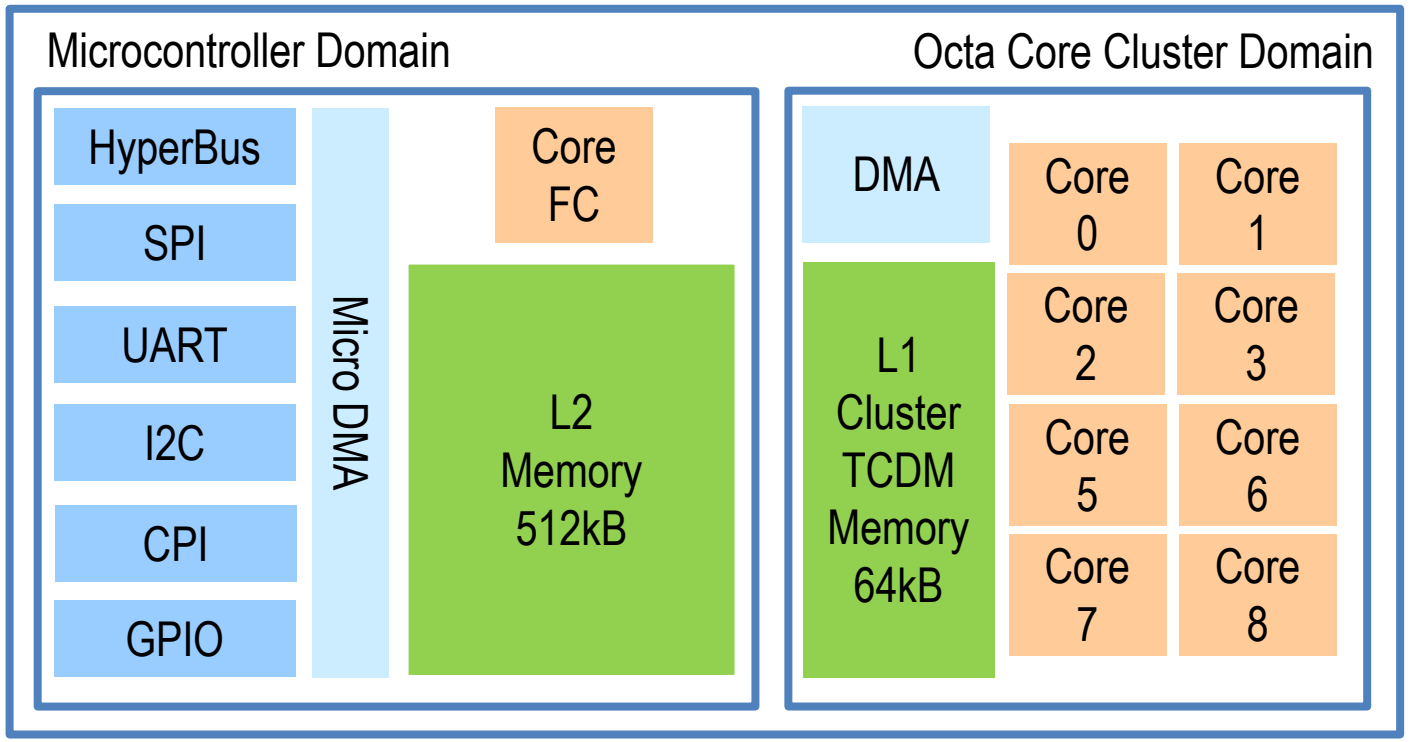
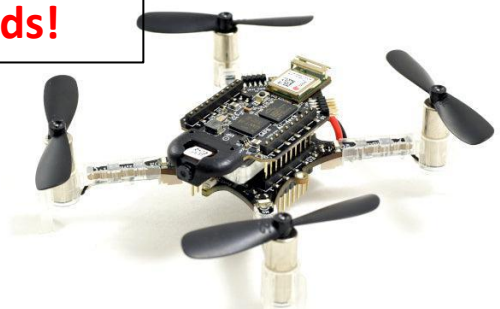
PULP Platform: today we focus on the 8-cores cluster



- **Cores**: 1 + 8
- **On-chip Memories**
  - A level 2 Memory, shared among all cores
  - A level 1 Memory, shared by the 8-cores cluster
- **cluster-DMA:** A multi-channel 1D/2D DMA, controlling the transactions between the L2 and L1 memories
- **micro-DMA**: A smart, lightweight and completely autonomous DMA () capable of handling complex I/O scheme
  - **Bus+Peripherals:** HyperBus, I2S, CPI, timers, SPI, GPIOs, etc…

NB: this is the architecture you find on our nano-drones and GAP boards!



GitHub HW Project: https://github.com/pulp-platform/pulp
HW Documentation: https://raw.githubusercontent.com/pulp-platform/pulp/master/doc/datasheet.pdf

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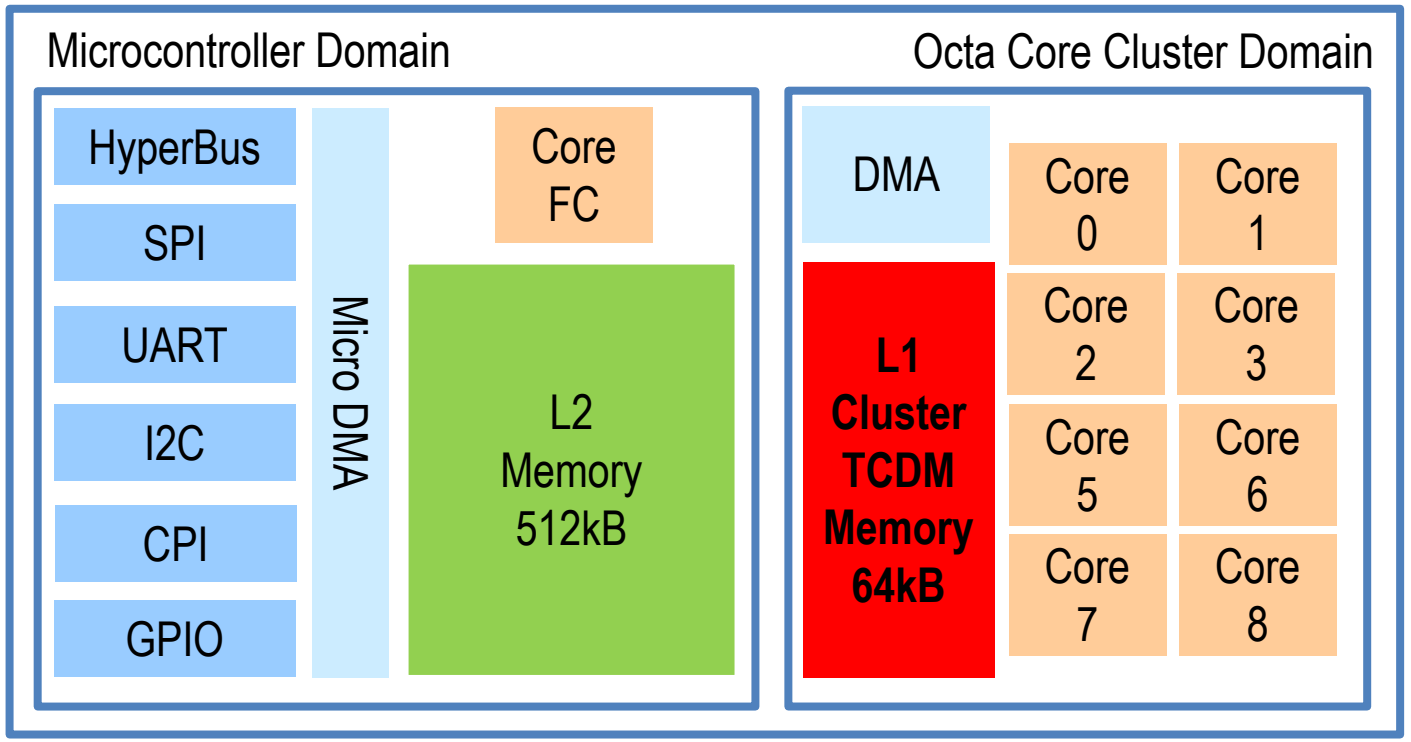

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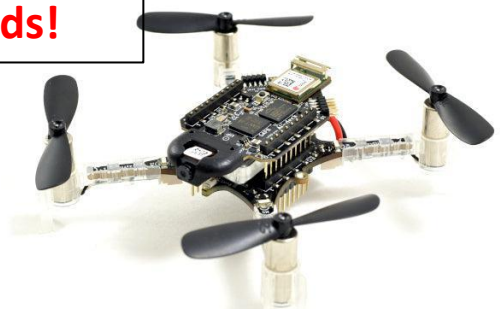

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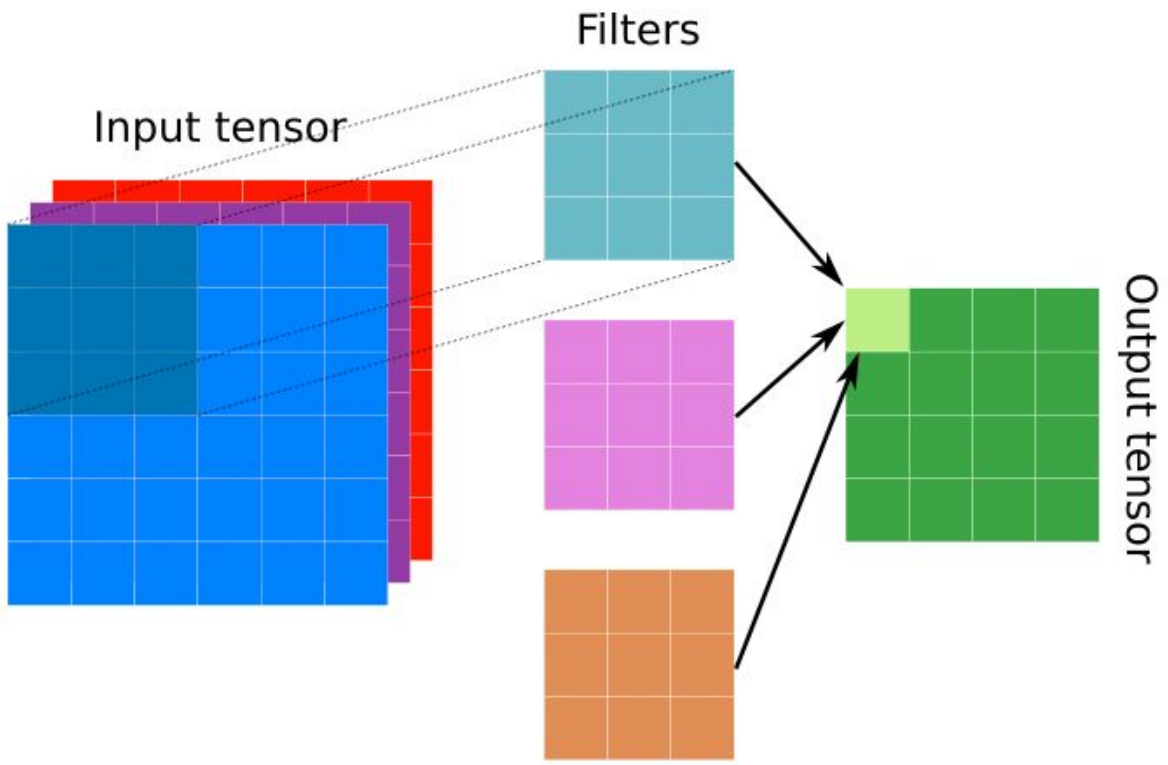
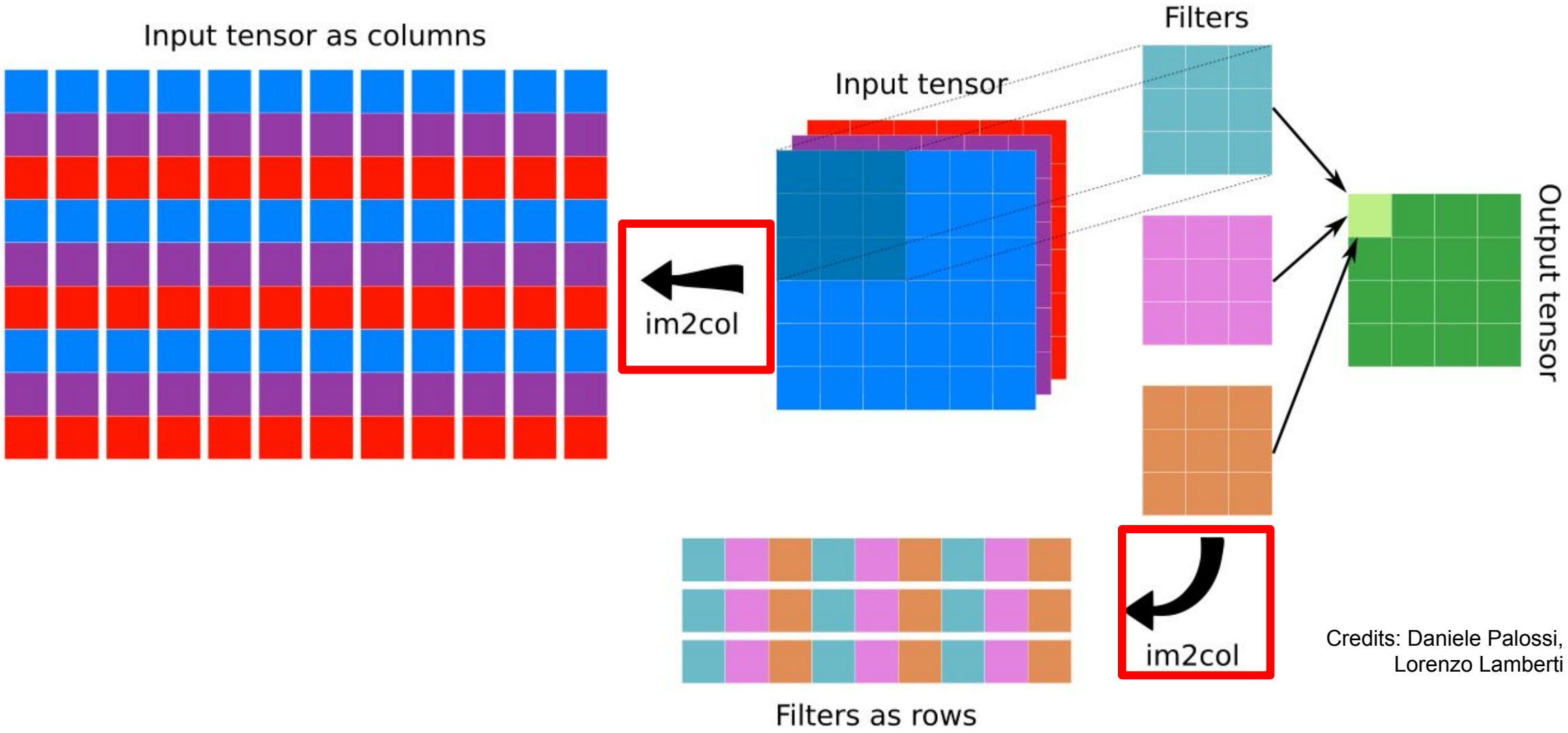
Convolution Operation: naive



Credits: Daniele Palossi,
Lorenzo Lamberti

Convolution Operation: im2col and MatMul



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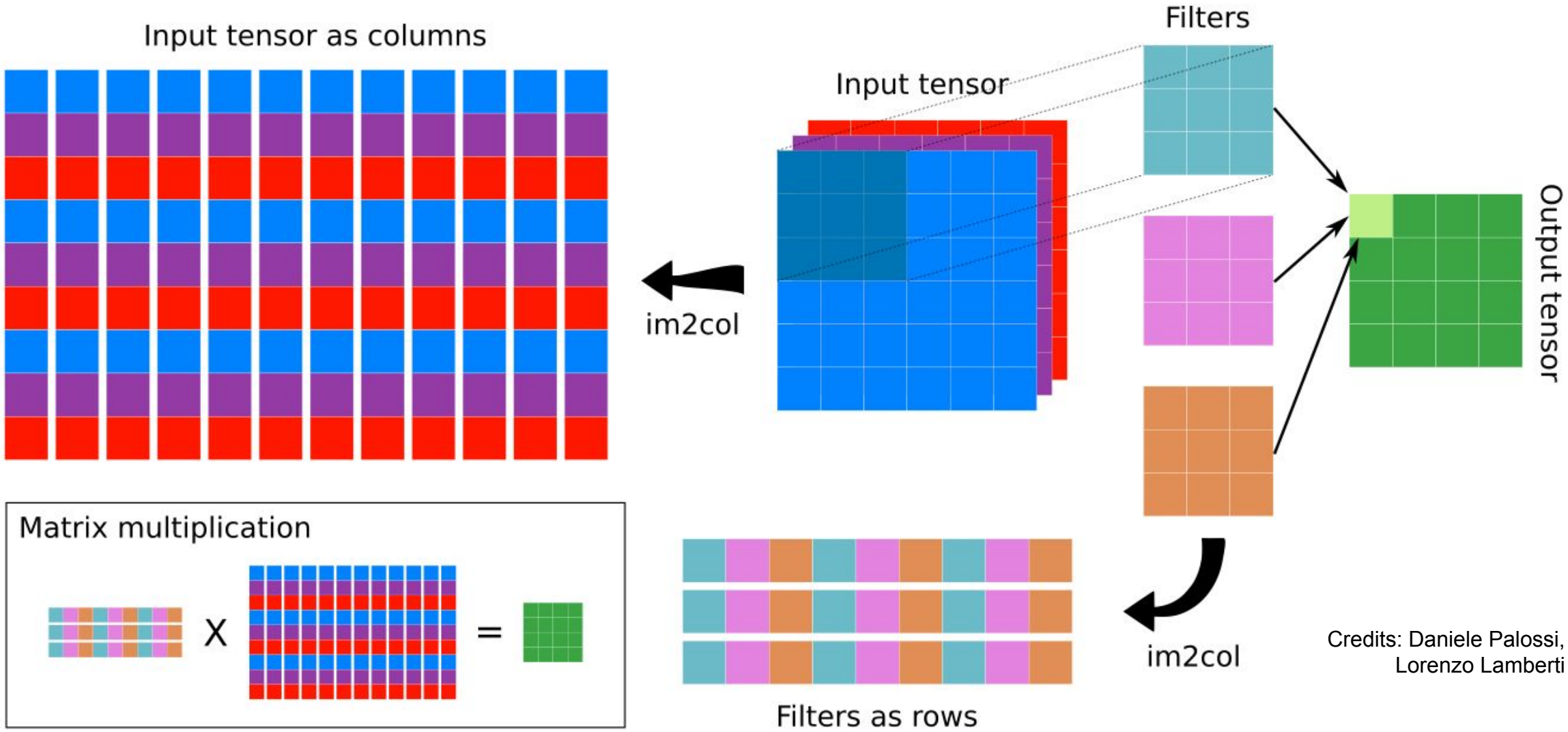

Credits: Daniele Palossi, Lorenzo Lamberti

EX1: find maximum dimensions of layers fitting L1 without tiling

```
Prerequisites:
source setup-pulp-sdk.sh

Run the code:
$ python3 parameters_generate.py --channels=<add_here> --spatial_dimension=<add_here>
$ make clean all run
```

Follow the assignment document.

NB: Choose the exercise by uncommenting one of the following defines in `Inc/main.h`:

Remember to change "CHANNELS" and "SPATIAL_DIM" **in Src/main.c** when you change layer size!

```
#define EXERCISE1
//#define EXERCISE2
// #define EXERCISE3
```


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Exercise 1

L1 memory: 64kB = ~~64000~~ (consider 50KB +/- 2KB as Maximum).

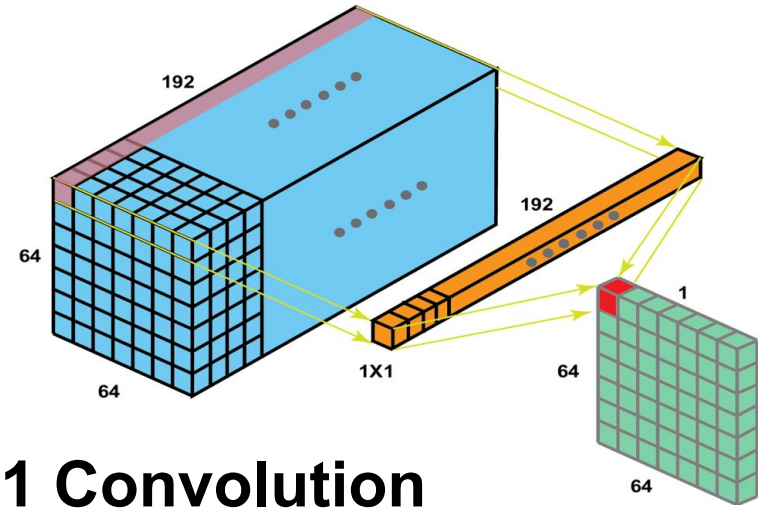
We must fit: input, kernel, output
Ch = 16
W, H = ?
Spatial dimension = W = H

Input size= W * H * Ch
Kernel_size= W_k * H_k * Ch_in * Ch_out = 1*1*16*16
Output_size= W * H * Ch
IM2COL size= 2 * 8 * W_k * H_k * Ch_in

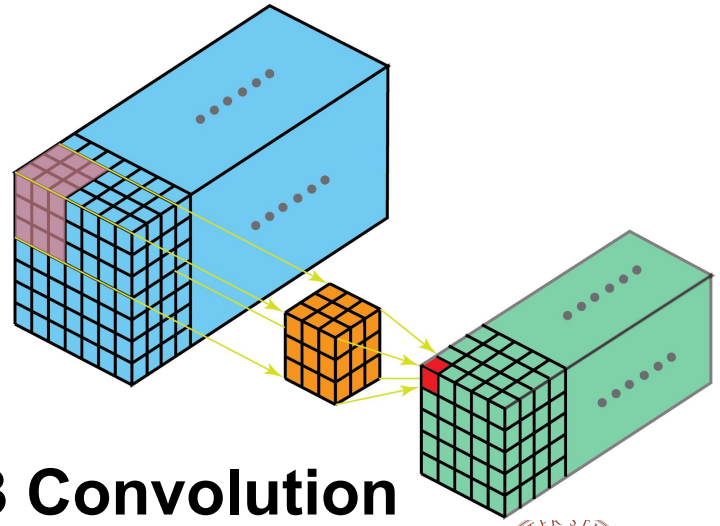
We want to solve this:
Input + kernel + output < L1

(W*H*16) + (16*16) + (W*H*16)=
(W^2*16) + (16*16) + (W^2*16) < 52000

16 W^2 + 256 + 16 W^2 < 52000
32 W^2 < 52000 - 256



1x1 Convolution
Used today!



3x3 Convolution
not used today

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TASK2: fetch from L2

EX2: fetch data from L2

```
Run the code:
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$ make clean all run
```

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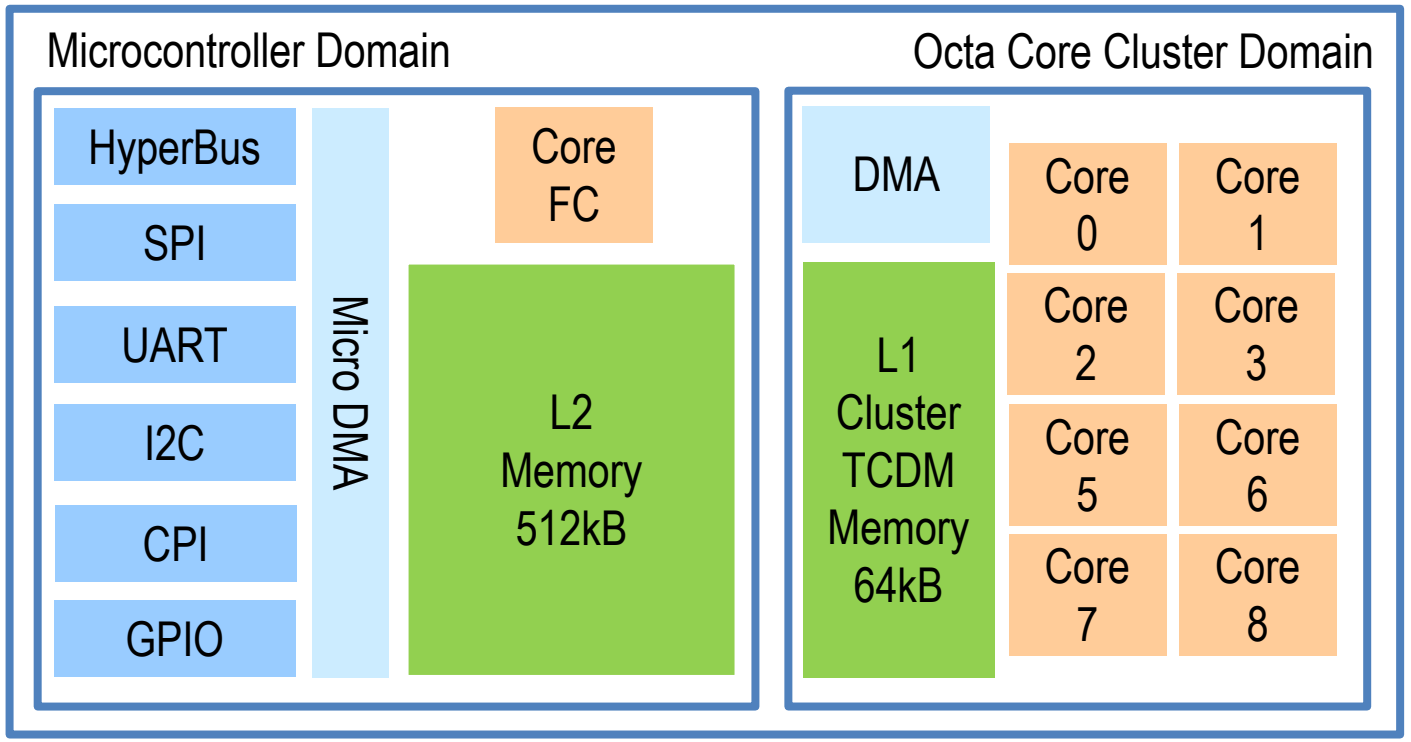

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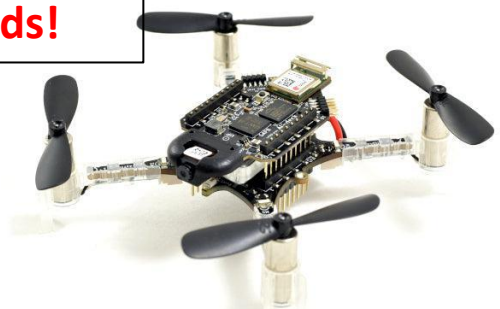

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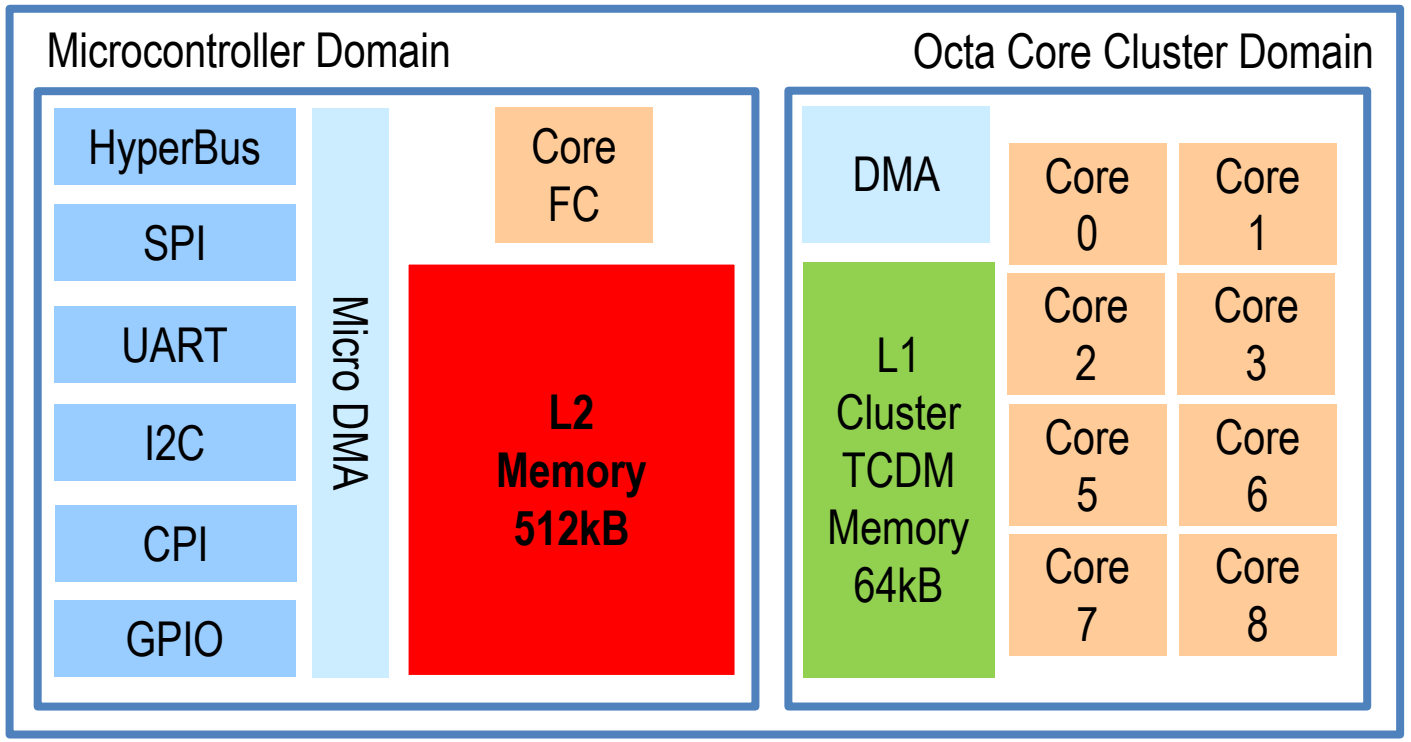

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Slower access from cluster!!

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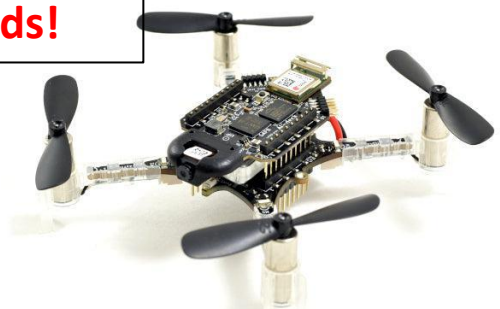

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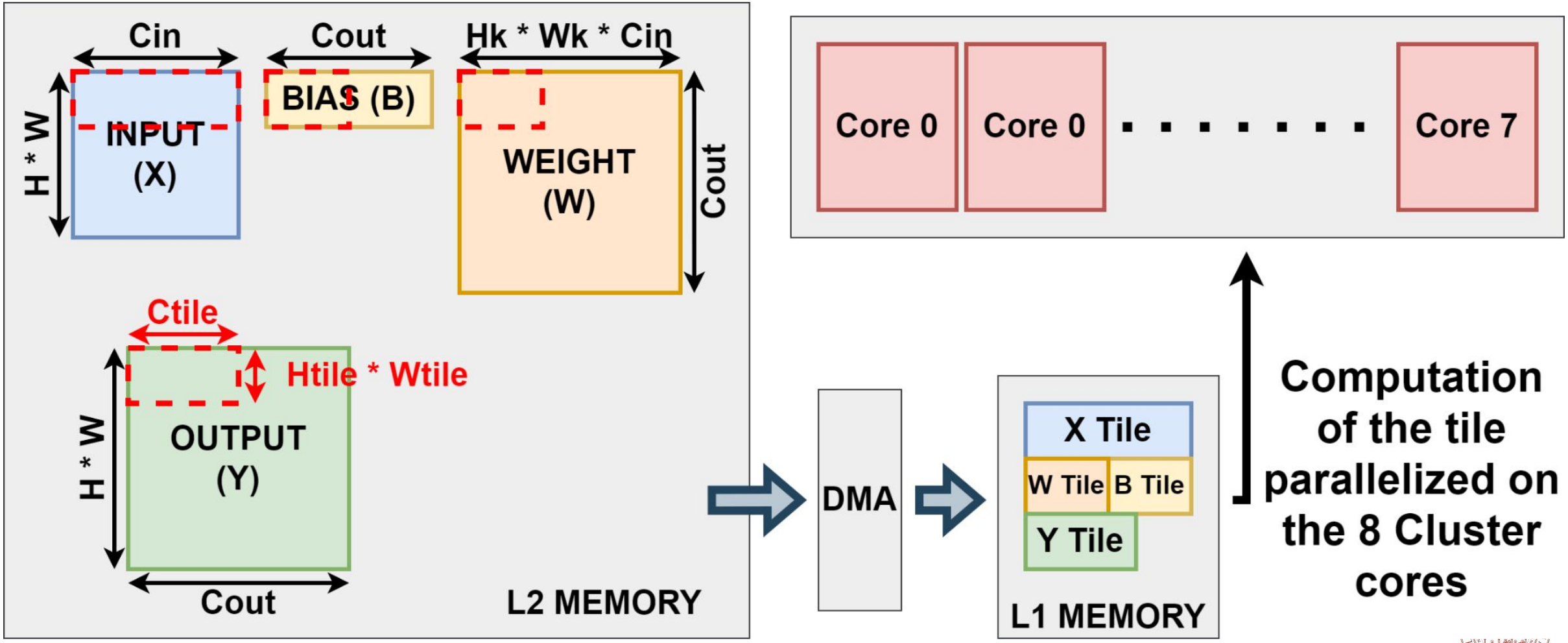
TASK3: Tiling

Tiling

Tiling: for each output chunk (tile), bring to L1 only the portion of the input/weight tensors necessary to compute it, then move the result to L2 an go on with the next tile!



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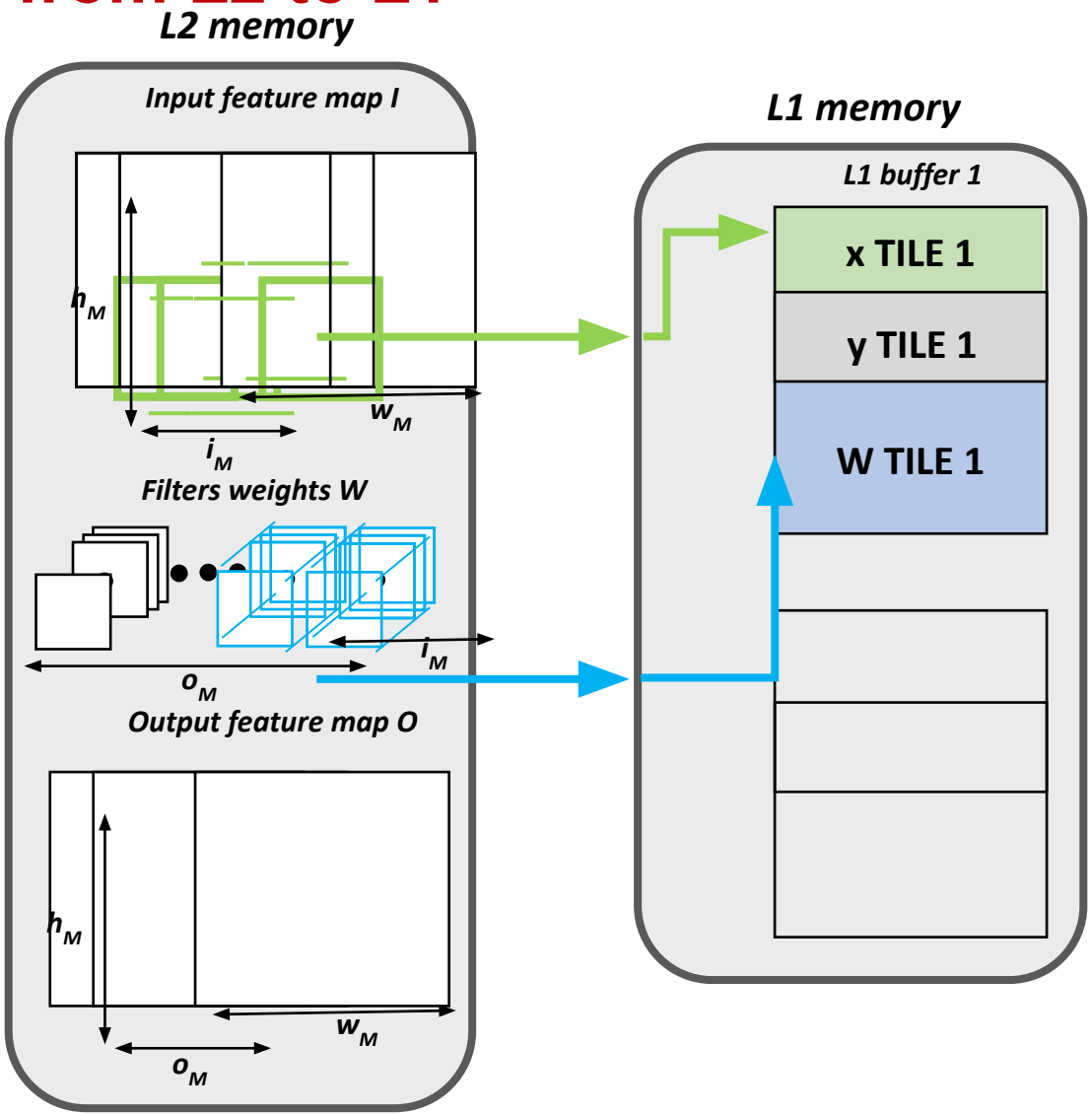
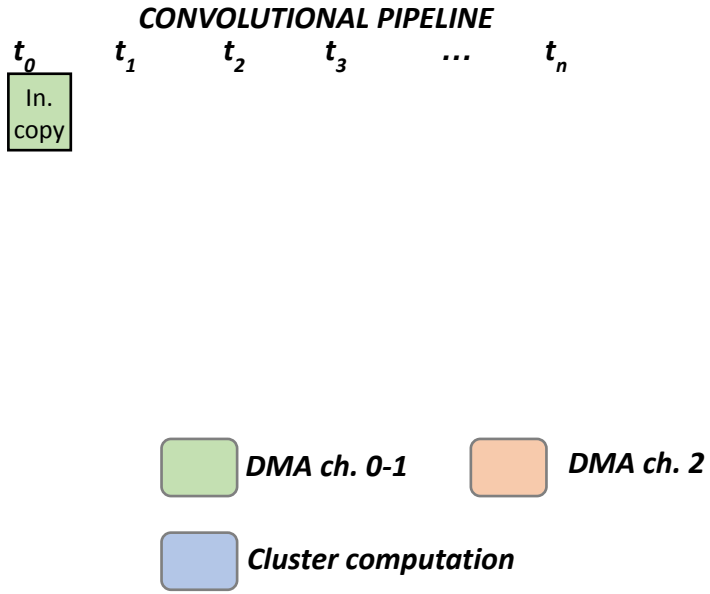
Tiling from L2 to L1





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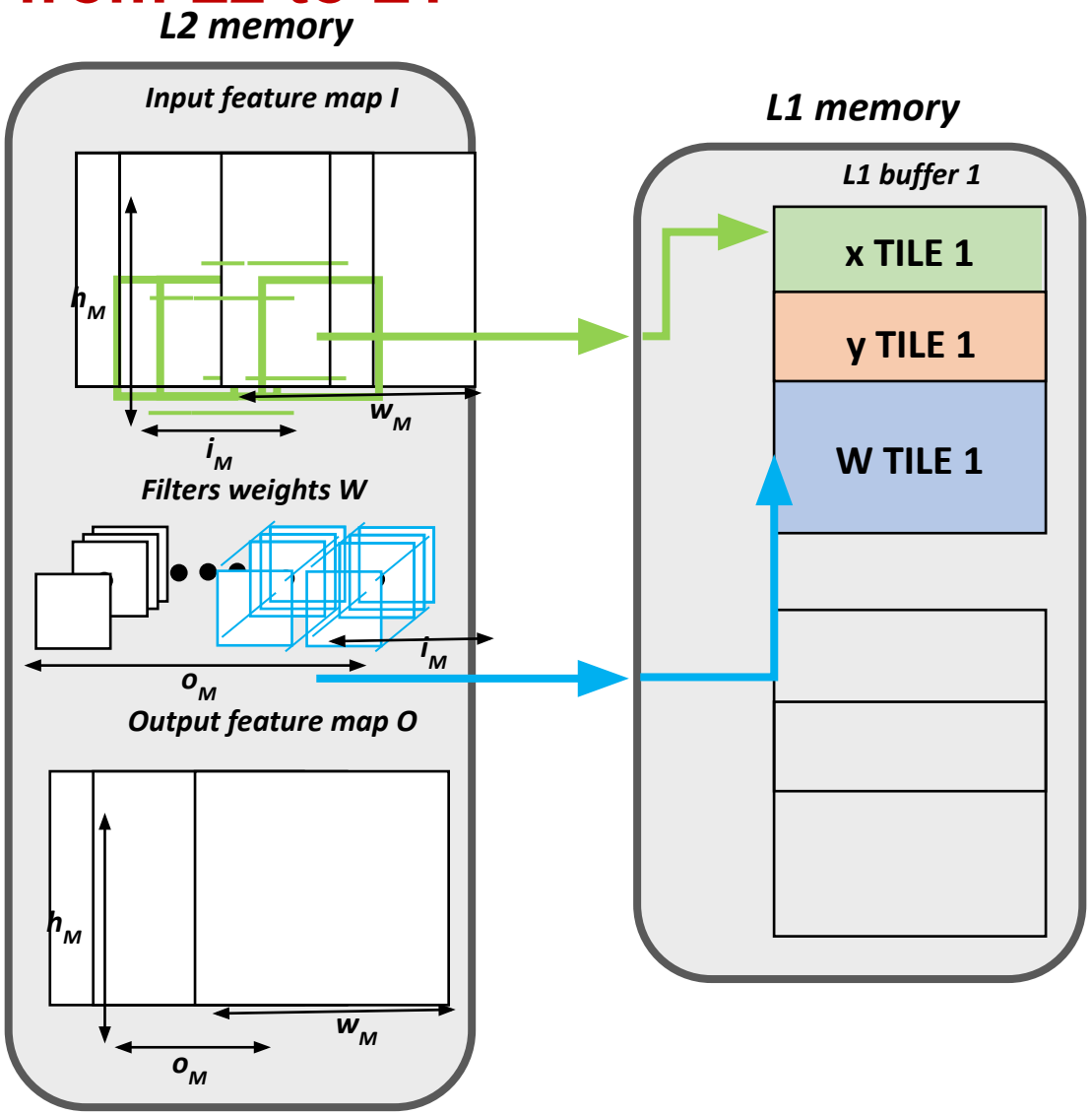
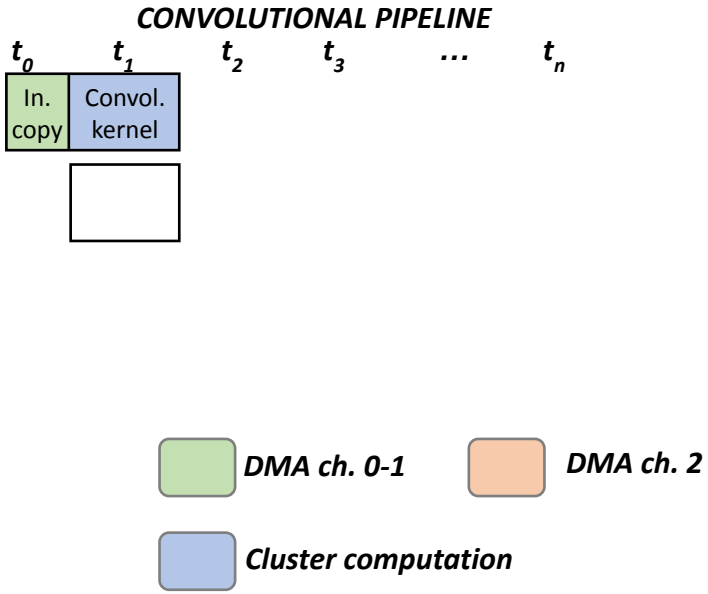
Tiling from L2 to L1





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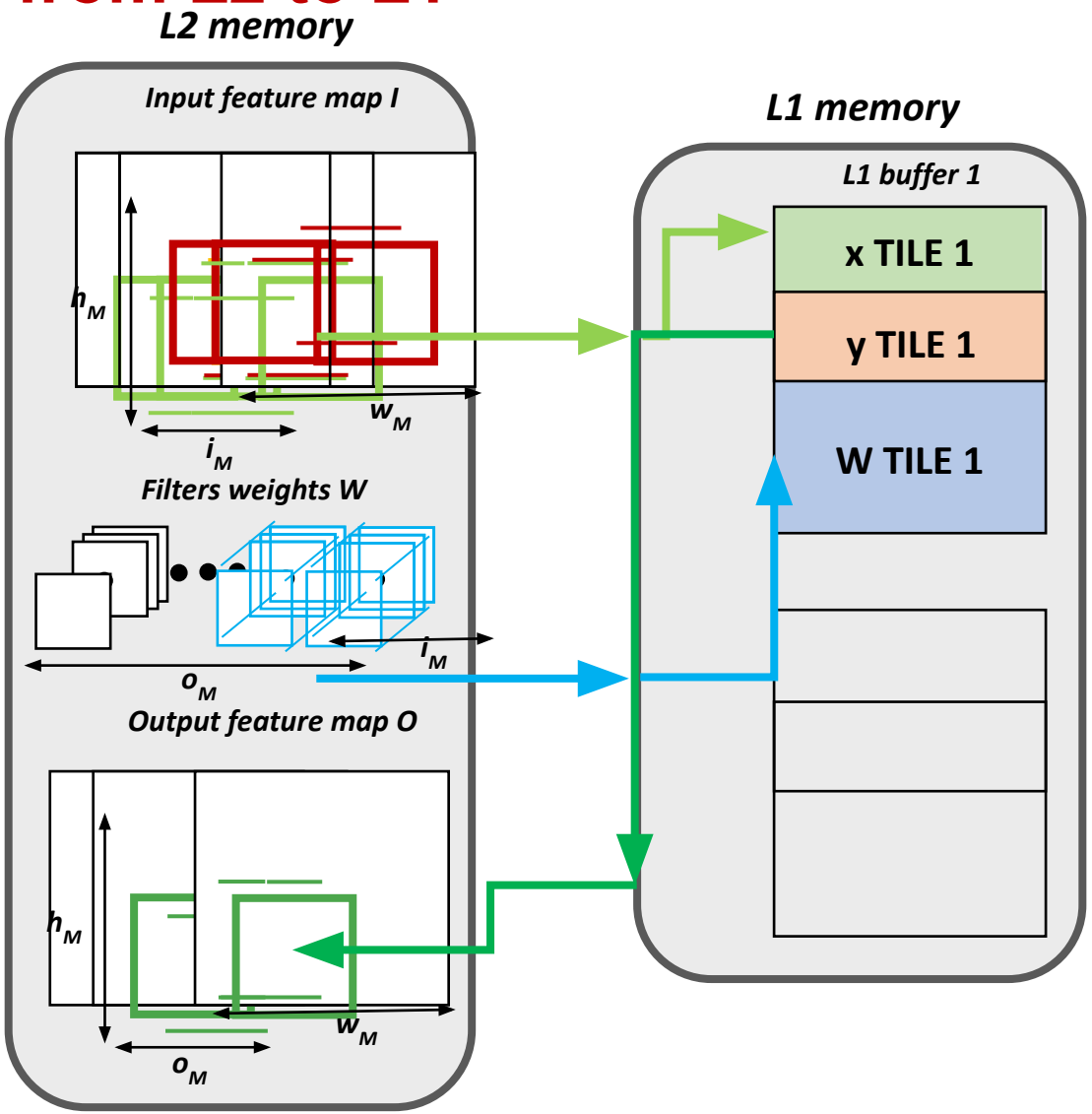
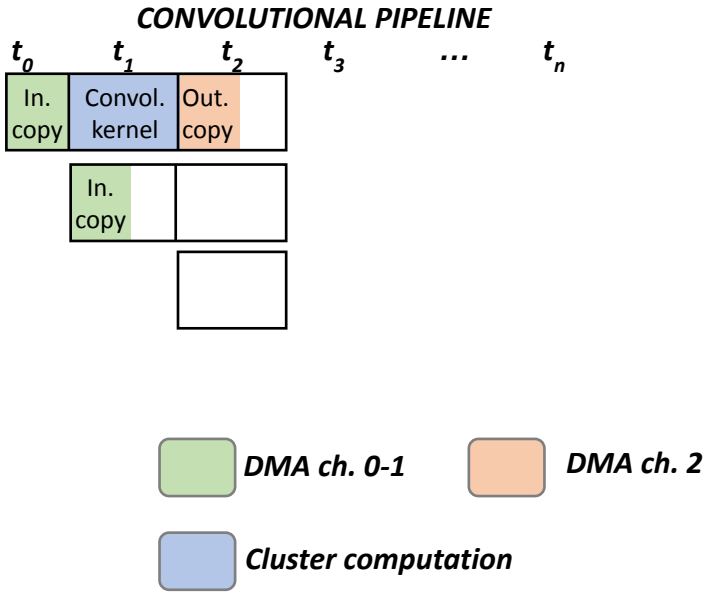
Tiling from L2 to L1





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EX3: Tiling layer

```
Run the code:
$ python3 parameters_generate.py --channels=#### --spatial_dimension=####
$ make clean all run
```

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